

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (VPA DD)

PUBLICATION DATE **14.04.2023**

VERSION **v. 2.1**

RELATED SUPPORT

[- Programme of Activity requirements](#)

[- TEMPLATE GUIDE VPA Design Document v.2.1](#)

This document contains the following Sections

Section A – Description of project

Section B – Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

Section C – Duration and crediting period

Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

Section E – Summary of Local stakeholder consultation

Section F – Eligibility and inclusion criteria for VPAs inclusion

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2- Contact information of VPA Implementer (mandatory)

Appendix 3 – LUF Additional Information (VPA specific)

Appendix 4 - Design Changes

KEY PROJECT INFORMATION

Type of VPA	☒ Real case VPA ☐ Regular VPA
Scale of VPA Note that a VPA can be of one scale. Please select applicable scale accordingly.	☐ Microscale ☐ Small scale ☒ Large scale
Title of corresponding real case VPA (if applicable)	GS 11142-Installation of High Efficiency Firewood Cookstoves in Sub-Saharan Africa Installation of High Efficiency Firewood Cookstove in Malawi-GS 11359
GS ID of real case VPA (if applicable)	11359
GS ID of VPA	11359
Title of VPA	GS 11142- Installation of High Efficiency Firewood Cookstoves in Sub-Saharan Africa VPA 2-Installation of High Efficiency Firewood Cookstoves in Malawi
Time of First Submission Date	09/09/2021
Date of Design Certification	NA
Version number of the VPA-DD	3.3
Completion date of version	24/08/2023
Coordinating/managing entity	C-Quest Capital Stoves Asia Limited
VPA Implementer (s)	NA
Project Participants and any communities involved	N/A
Host Country (ies)	Republic of Malawi
GS ID and Title of applicable Design Certified VPA	N/A
GS ID and Title of applicable Performance Certified VPA	N/A
Activity Requirements applied	☒ Community Services Activities

	☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Other Requirements applied	Usage Rate Requirements, Version 2.0
Methodology (ies) applied and version number	Technologies and Practices to Displace Decentralized Thermal Energy Consumption (V3.1)
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
VPA Cycle:	☒ Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

SUSTAINABLE DEVELOPMENT GOALS TARGETED	SDG IMPACT (DEFINED IN B.6.)	ESTIMATED ANNUAL AVERAGE	UNITS OR PRODUCTS
13 Climate Action (mandatory)	Emission reductions	147,004	VER (tons CO2/annum)
15 Life on Land	Amount of firewood saved by all the project stoves per year	88,435	Biomass in tons
3 Good health and well-being	Reduced incidence of disease caused by air pollutants.	29,700	Number of households
5 Gender Equality	Average time saving associated with cooking time and fuel collection	209	Hours of drudgery saved per HH/year
4 Quality Education	Skill development	20	Number of skilled individuals

8 Decent Work and Economic Growth	Increased employment opportunities	50	Number of staff
7 Affordable and clean cooking	Energy Efficiency improvement	1,558	Biomass Energy saved in TJ/yr
1 No Poverty	Household saving	19,995	Kwacha/year/household Which was purchasing firewood in the baseline scenario

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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C-Quest Capital plans to install 50,000 domestic high efficiency cookstoves in total in rural Malawi where 90.9% of the households rely on firewood as main source of energy for cooking¹. Fuelwood consumption has been increasing during last two decades and has doubled from 21 million m³ in 2000 to 41 million m³ in 2018². On the other hand, forest cover in Malawi has been constantly dropping with an increasing rate during the last 30 years. 42 thousand ha of forest land has been lost annually during this period. (Global Forest Resources Assessment 2020- Main Report³).

The project stoves disseminated will replace the prevailing inefficient three stone fires or traditional pot supports with stoves which combust wood more efficiently and improve thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions. The ICL envisaged to be implemented under the VPA is TLC Rocket Stove, which results in substantially reduced fuel consumption and emissions for conducting cooking and water heating tasks in homes.

The project plans to install improved cookstoves to the households in rural regions of Malawi and Gold Standard approved Community Services Activity Requirements is applicable for the project. The project type is, therefore; automatically eligible as per section 4.1.3 of GS4GG Principles & Requirements.

Republic of Malawi is listed as Least Developed Country by UN and does not have an emission cap enforced. There is no policies enforced/under development for trading emission under the scope of the proposed project.

The project is compliance with all applicable legal, environmental, ecological and social regulations in Malawi. Improved cooking devices to include improved biomass stove,

¹ http://www.nsomalawi.mw/images/stories/data_on_line/economics/ihs/IHS5/IHS5_Final_Report.pdf

² <http://data.un.org/Data.aspx?d=EDATA&f=cmID%3aFW%3btrID%3a1231>

³ <http://www.fao.org/3/ca9825en/ca9825en.pdf>

use of ethanol and LPG stoves, and switch to electric stoves is considered as a mitigation measure under the Sustainable Forest Management programme in Intended Nationally Determined Contributions⁴. The implementation is said to be subject to the availability of international support in form of finance, technology and capacity building, which is now provided by carbon finance.

A.1.1. Eligibility of the VPA under approved PoA

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Table 2 Eligibility for VPA inclusion as per PoA requirements

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION AND SUPPORTING EVIDENCE FOR INCLUSION
1	Geographic boundary	Each VPA shall be located within the boundaries of the PoA.	Recorded geographical location in the project database to be checked by VVB
2	Double Counting	A unique numbering system will be adopted for each VPA. All ICS installed shall receive a unique number identifying the VPA to which it belongs.	Each ICS included in the VPA will have a unique combination of customer name and geographical location linked with a unique serial number. The Project Developer commits to retiring eligible units equal to the quantity of GS VERs if any double counting exists.

⁴

<https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Malawi%20First/MALAWI%20INDC%20SUBMITTED%20TO%20UNFCCC%20REV.pdf>

3	Exclusiveness of VPA	VPA shall not be previously: 1. Registered as a project activity with other offset schemes 2. Included as a VPA in any other registered PoA, or deregistered as a VPA of a PoA.	Confirmation by CME
4	Specification of Technology/ Measure	VPAs under the PoA shall install portable or in-situ ICS using firewood. The capacity is limited by 150kW as per the applied methodology Technologies and Practices to Displace Decentralized Thermal Energy Consumption version 3.1.	Each stove is assumed to have a thermal capacity of 1.61 kW. The real capacity will be determined by Kitchen Performance Test (KPT) and Water Boiling Test (WBT) results.
5	Start date	The start date of any proposed VPA will be on or after the start date of PoA.	The date on which first ICS installed will be recorded. Sales invoice or end-user agreement will be provided as a proof. The expected start date of the VPA is 01/07/2023 which is the date of installation of first stove included under it.
6	Applicability of methodologies	VPAs shall apply Technologies and Practices to Displace Decentralized Thermal Energy Consumption (v 3.1).	Please refer to section B.2 of VPA-DD for confirmation.
7	Additionality	The additionality PoA is demonstrated as per GS4GG Community services activity requirements, Version 1.2	Confirmation by the CME that the project is located in Malawi to be provided to the VVB. Total expected thermal energy savings per unit is 0.009634 GWh. Actual values shall be submitted to VVB at the time of verification.

8	Public funding	Affirmation that there is no diversion of Official Development Assistance (ODA).	Declaration of CME
9	Target group& Distribution Mechanism	Households will be the target group for VPAs. ICSs will be installed by local VPA Implementers on a non-commercial basis.	CME declaration or user registration. As the stoves included in the present VPA are mud-brick fixed stoves, the distribution mechanism adopted is house to house installations. The CME and/or its implementing partners carry out the stove building and installation with the help of beneficiaries. Post installation, the households are visited by the CME or IP team to assess whether the stoves are built properly and have dried and are ready to use. The registration is done during these visits, when it is deemed that the end users can start using the stoves.
10	Sampling requirements	All requirements as mentioned in Technologies and Practices to Displace Decentralized Energy Consumption, version 3.1 or the Standard: Sampling and surveys for CDM project activities and programme of activities are applicable to VPAs.	Please refer to Section B.7.2 Sampling Plan for details. VPA Implementers will follow the management system described at the PoA-DD.

11	Compliance of the technology implemented	The capacity of each ICS will comply with the requirements of applied methodology- Technologies and Practices to Displace Decentralized Energy Consumption.	VPA applies GS methodology- Technologies and Practices to Displace Decentralized Energy Consumption (v3.1) Please refer to Section B.2 of the VPA-DD for demonstration of its compliance with TPDDTEC requirements.
12	SDG Outcomes	Each VPA shall conduct SDG outcomes assessment and comply with the SDG targets identified in the PoA-DD.	Each VPA shall conduct SDG outcomes assessment and comply with the SDG targets identified in the PoA-DD. Please see Stakeholder Consultation Reports for details.
13	Stakeholder Consultation	Each VPA will conduct a local stakeholders' consultation.	Local Stakeholder Consultation Meeting is conducted on July 29 th 2021. Please see Stakeholder Consultation Report for details.
14	VER Ownership	End users receiving ICSs under the specific VPA contractually cede their rights to claim and own emission reductions to the CME of the PoA.	Carbon right waiver agreement between CME and the user. A sample carbon waiver has been provided to VVB.

A.1.2. Legal ownership of products generated by the VPA and legal rights to alter use of resources required to service the project

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The Verified Carbon Reductions (VER) generated by each stove belongs to the individual stove user. The purchase agreement entered between the stove user and the stove producers includes terms transferring the ownership of VERs from user to the producers. By the purchase of the stove, the user accepts to waive the carbon rights.

A.2. Location of VPA

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Republic of Malawi (Figure 1)



Figure 1

Coordinates for Malawi: 13.2543° S, 34.3015° E

A.3. Technologies and/or measures

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The TLC Rocket Stove is a simple design with basic features. The design uses bricks that are made by the household using locally available clay. The average size of the brick used on the TLC Rocket Stove which is produced using a standard mould is 22cm (L) x 11cm (W) x 6.5cm (H). The bricks are mortared together using locally available material (clay soil, cow dung, and sand) for better insulation and heat loss reduction. The mud mortar is a mix of 5 liters clay, 5 liters sand, 5 liters manure with 5 liters of water. Average lifespan for the cookstoves is expected to be 10 years. Figure 2 below shows how the cook stove is assembled.

Table 3. Technical Specifications

Technical Specifications – TLC Rocket Stove

Thermal efficiency	34.5 % ⁵
Size (stove)	Depth: 34.5 cm Width: 34.5 cm Height: 28 cm
Size (combustion chamber)	Depth: 12 cm Width: 12 cm Height: 28 cm

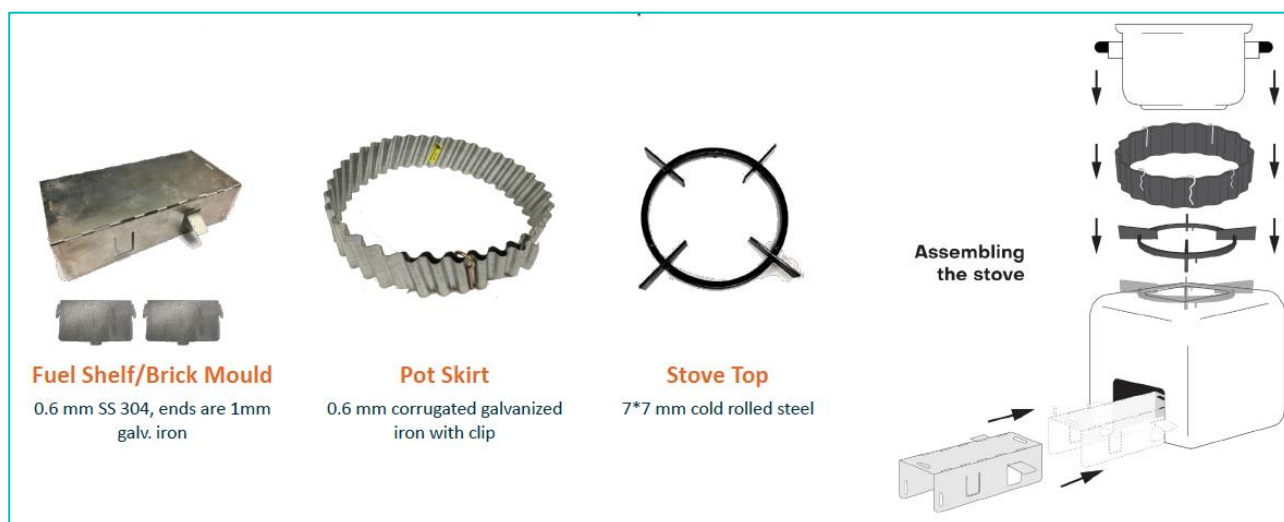


Figure 2. Parts of the project stove

The stainless-steel stick shelf allows air to flow underneath the wood sticks into the combustion chamber resulting in cleaner and more efficient combustion.

The adjustable corrugated steel pot skirt improves the transfer of heat from the fire into the pot, increasing overall energy efficiency of the stove.

A.4. Scale of the VPA

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The project is large scale based on project scale defined under Section 9.1.2(b) in GHG Emissions Reduction & Sequestration Product Requirements (v2.0). The aggregate energy savings of the project activity exceeds the limit of small scale, equivalent of 180 GWh thermal per year in fuel input. Each ICS installed saves 0.009634 GWh and 50,000

⁵ Based on the WBT results of TLC 15 Brick Rocket Stove, October 2015, Aprovecho Research Center

stoves are planned to be distributed; which is 481.7 GWh thermal energy savings per year. Thermal efficiency savings per stove is calculated as follows:

$$TE_{savings} = B_{old} \times \left(1 - \frac{n_{old}}{n_{new}}\right) \times NCV_{biomass} \times f_{GWh}$$

$$TE_{savings} = 3.26 \frac{\text{ton}}{\text{y}} \times \left(1 - \frac{0.1}{0.345}\right) \times 0.015 \frac{\text{TJ}}{\text{ton}} \times \frac{0.27778 \text{GWh}}{\text{TJ}} = 0.009634 \text{ GWh}$$

$TE_{savings}$: Thermal Energy savings per project stove (GWh)

B_{old} : Annual fuel consumption in baseline scenario per unit per year (ton/y)

n_{old} : Default value for thermal efficiency of the three stone baseline stove (%)

n_{new} : Thermal efficiency of the project stove as per WBT results (%)

$NCV_{biomass}$: Net calorific value for firewood (TJ/ton)

f_{GWh} : Conversion factor for TJ to GWh

A.5. Funding sources of VPA

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The project will be implemented by C-Quest Capital; the producer of the improved cookstoves and Project Developer, in cooperation with the local partners or Project Implementers. The funding for the production of stoves will be provided by C-Quest Capital that will be legal owner of the VERs. No public funding or Official Development Assistance is involved. The ODA Declaration signed by the CME will be provided as per GS4GG requirements.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

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GS Methodology: Technologies and Practices to Displace Decentralized Thermal Energy Consumption" (TPDDTEC) Version 3.1 (August 2017)

Applicable GS4GG requirements are as follows:

- Principles and Requirements (V1.2)
- GHG Emissions Reduction & Sequestration Product Requirements (v2.1)
- Community Services Activity Requirements (v1.2)
- Usage Rate Requirements (27/10/2020)

B.2. Applicability of methodology (ies)

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The methodology- Technologies and Practices to Displace Decentralized Thermal Energy Consumption (V3.1) methodology defines the following applicability criteria:

Applicability	Justification
The project boundary needs to be clearly identified, and the technologies counted in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases, there may be another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The project boundary will be within the territorial limits of Malawi. Each stove installed under the VPA shall be identified through a combination of unique serial number and GPS coordinates enabling their identification with the VPA. Prior to each verification, the project database will be checked to make sure that only stoves belonging to the VPA are included for emission reduction calculation as well as stoves included in this VPA are not included in any other GHG program or VPAs belonging to this particular PoA. The registration card with corresponding stove serial number will ensure that no double counting of stoves takes place.

	If stoves are transferred from other GHG programme to the said VPA, a zero-issuance certificate from the programme registry shall be submitted to the verifying VVB to ensure that credits are not being claimed under GS and other programme for same vintage.
The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The capacity of each stove is calculated as 1.61 kW to achieve the calculated energy savings⁶. That is only 1% of the 150kW limit. Please refer to the calculations provided below this table.
Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear	Continued use of baseline stoves will be monitored during annual/biennial surveys and the emissions related with the use will be accounted in the overall emission reduction amount.

⁶ The value is based on the WBT results of TLC 15 Brick Rocket Stove, October 2015, Aprovecho Research Center. The real value will be determined by the results of field performance tests (WBT and KPT).

description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in use after the introduction of the improved technology.	
The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in use, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users will need to be informed and notified that they cannot claim for emission reductions from the project	Each user will be informed about the transfer of carbon rights and an agreement will be signed between Project Developer and the user. How the project generates carbon credits and what kind of project activities will be financed by the revenue from carbon credits are discussed in detail during the Stakeholder Consultation Meeting as well.
Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules. Such projects should further fulfill the following conditions a. Air pollution (IAP) levels are not worsened compared to the baseline, and greenhouse gases emitted by the	N/A

project fuel/stove combination are estimated with adequate precision. b. Records of renewable fuel sales may not be used as sole parameters for emission reduction calculation	
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Thermal capacity of the project stove is calculated as follows:

Step 1: Estimate the energy consumption in kWh

$$B_{new} = B_{old,i,j} \times \left(\frac{n_{old}}{n_{new}} \right) = 3.26 \frac{\text{ton}}{y} \times \left(\frac{0.1}{0.345} \right) = 0.9 \frac{\text{ton}}{y} \rightarrow 0.0026 \frac{\text{ton}}{\text{day}}$$

$$TE = 0.0026 \frac{\text{ton}}{\text{d}} \times 0.015 \frac{\text{TJ}}{\text{ton}} \times 277,778 \frac{\text{kWh}}{\text{TJ}} = 10.7730 \text{ kWh/day}$$

Step 2: Estimate energy output:

Based on a survey done by World Bank for Sub-Saharan countries, households spend 3 hours on average for cooking⁷. Accounting that 77%⁸ of cooking is done with the project stove, number of hours for the project stove is calculated as 2.31 hrs in a day. Therefore; the capacity of stove is 10.7730 kWh/2.31h= 4.6636 kW.

Step 3: Estimate the useful thermal energy output:

Assuming the thermal efficiency of the stove is 34.5%; the useful output will be 4.6636 kW *0.345= 1.61 kW.

B.3. VPA boundary

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The Project boundary is physical, geographical site of individual improved cookstoves that utilize biomass.

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<https://openknowledge.worldbank.org/bitstream/handle/10986/22521/Clean0and0impr000a0landscape0report.pdf?sequence=1&isAllowed=y> (Page:154)

⁸Monitoring report dated 13/08/3019 for CPAs of CDM PoA8060 states that 23% of the households uses a second stove at least once per week as per the baseline study conducted.

Source	GHGs	Included?	Justification/Explanation	
Baseline	Source 1	CO ₂	Yes	Major source of emissions
	Combustion of nonrenewable biomass	CH ₄	Yes	Important source of emissions
	for cooking with traditional methods	N ₂ O	Yes	Can be significant source of emissions for some fuels
Project	Source 1	CO ₂	Yes	Major source of emissions
	Combustion of nonrenewable biomass	CH ₄	Yes	Important source of emissions
	for cooking with ICS	N ₂ O	Yes	Can be significant source of emissions for some fuels

B.4. Establishment and description of baseline scenario

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As per the applied methodology, the project proponent must conduct the following baseline studies for each baseline scenario:

- Baseline non-renewable biomass (NRB) assessment, if biomass is one of the baseline fuels
- Baseline survey (BS) of target population characteristics
- Baseline performance field test (BFT) of fuel consumption (e.g. kitchen performance test (KPT) in case of cook stoves)

Findings of the performance field tests will be submitted prior to first verification. A project estimation of expected baseline emissions is provided for listing of the project using available sources of information.

Baseline non-renewable biomass (NRB) assessment has been calculated by an independent study carried out as per "CDM Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0. The fNRB value for Malawi is calculated to 0.85 and will be used in emission reduction calculations.

Baseline survey (BS) of target population characteristics

The baseline survey provides critical information on target population characteristics, baseline technology use, fuel consumption, leakage, and sustainable development indicators.

Baseline Survey Representativeness

The baseline survey requires in person interviews with a robust sample of end users without project technologies that are representative of end users targeted in the project activity.

Rural households in Malawi using fuelwood for cooking are the target population.

The annual fuelwood consumption is calculated as 3.26 t/hh/yr as per the previous Baseline Firewood Consumption Study done by HED Consulting in December, 11th 2012. The actual value of fuelwood consumption in the project stove will come from KPTs conducted on baseline and project stove. The value of 3.26 t/hh/yr has been used only for ex-ant calculation.

Baseline Survey Sample Sizing

The baseline survey should be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller
- Group size 300 to 1000: Minimum sample size 10% of group size
- Group size > 1000: Minimum sample size 100

More than 100 households will be sampled using simple random sampling.

Data to be collected

The data collected is specific to the characteristics of each baseline scenario, and should be tailored accordingly. Information on the following needs to be gathered:

1. User follow up
 - a. Address or location
 - b. Mobile telephone number and/or landline telephone number (when possible)
2. End user characteristics
 - a. Number of people served by baseline technology
3. Baseline technology and fuels
 - a. Types of baseline technologies used and estimated frequency
 - b. Types of fuels used and estimated quantities
 - c. Seasonal variations in baseline technology and fuel use
 - d. Sources of fuels; (purchased or -collected, etc) and prices paid or

effort made (e.g. walking distances, persons collecting, opportunity cost)

e. Renewability and non-renewability indicators as required by Annex 1

The survey included questions about basic household characteristics including the number of people living in the household, the number of meals cooked per day, education, source of income, stove and fuel use before and after installation of the project stove, and season variations of stove and fuel use patterns. Contact details of the participants are also recorded.

Results and analysis

- 86% of the households were found using three-stone fire while 2% reported using traditional pot support/other conventional stoves along with TSF. Interestingly, 11% of the households in baseline reported using Chitetezo Mbawula stove as their primary cookstove.
- Maximum number of households (72%) cook 3 meals per day averaging to 21 meals in a week.
- Almost all respondents reported collecting firewood to meet their cooking energy needs and approximately 88% of the wood was sourced through collection. 31% of households purchased firewood and average quantity reported to be purchased was 12.5% of their requirement. There was just 1 household out of 100 which reported purchasing 100% of firewood used in cooking.
- 80% of the households said that they used a separate enclosed kitchen for cooking while 20% cook inside the house.
- 98% of the respondents surveyed, reported health issues owing to using their current cookstove. For households cooking inside the house, this number was 100%.

For households that collect 100% of their fuelwood:

- The fuelwood was collected around the village, forest and own land.
- In most of these households, female adult was responsible for fuelwood collection.
- The average trip made for collecting firewood was 3 per week and average round trip distance travelled for this purpose was 2 kilometers per trip. Women end up spending as much as approximately 2 hours per trip and an average of 6 hours per week in collecting firewood.

Baseline performance field test (BFT) of fuel consumption

As per the methodology, the baseline Kitchen Performance Test (KPT) should include a sample of end-users without project technologies that are representative of the end users targeted in the project activity. The baseline survey and baseline KPT can be conducted concurrently with the same end users. Any sampling methods can be used.

In cases of paired and independent sampling, there are two valid options for the statistical analysis:

- 90/30 rule: When the sample size is large enough to satisfy the 90/30 rule, i.e. the endpoints of the 90% confidence interval lie within +/- 30% of the estimated mean, overall emission reductions can be calculated on the basis of the estimated MEAN annual emission reduction per unit or MEAN fuel annual savings per unit.
- 90% confidence rule (Lower bound of the one-sided 90% confidence interval). When the result obtained does not satisfy 90/30 precision, the emission or fuel saving result is not the mean (or average) test result, but a lower value, i.e. the LOWER BOUND of the one-sided 90% confidence interval.

Paired sampling for KPT – sampling the same users before and after beginning use of a new cookstove – will be performed. Among the over 100 end users in the baseline survey, 30 households with similar socioeconomic and demographic characteristics with targeted population will be selected to perform baseline KPTs. The households did not use the project stove prior to the tests. Each will be provided with a stove afterwards.

B.5. Demonstration of additionality

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Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).

Activity Requirement for Community based technologies:

4.1.9. Projects that meet any of the following criteria are considered as deemed additional and therefore are not

required to prove Financial Additionality at the time of Design Certification:

- (a) Positive list (Annex B)
- (b) Projects located in LDC, SIDS, LLDC
- (c) Micro-scale projects

Describe how the proposed VPA meets the criteria for deemed additionality.

The proposed project is located in the Republic of Malawi which falls under the category of LDC and LLDC. The project also complies with the requirement 1.1.3 of Annex B: Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh ($1.8 \text{ GWh}_{\text{th}}$) of energy savings per year or ≤ 600 tonnes of emission reductions per year. Each stove is assumed to be save 0.009634 GWh and $3.34 \text{ tCO}_2\text{e}$ per year.

B.5.1. Prior Consideration

The project is applying under regular project cycle and demonstration of prior consideration is not required.

B.5.2. Ongoing Financial Need

>>N/A

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

SUSTAINABLE DEVELOPMENT GOALS TARGETED	MOST RELEVANT SDG TARGET	SDG IMPACT
		INDICATOR (PROPOSED OR SDG INDICATOR)
13 Climate Action (mandatory)	13.b. Promote mechanisms for raising capacity for effective climate change-related planning and management in least developed countries and small island developing States, including focusing on women, youth and local and marginalized communities	Emission reductions achieved by fuelwood savings at household level
15 Life on Land	15.2. By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.	Amount of non-renewable firewood saved by the project stoves per year
3 Good health and well-being	3.9. By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Number of households that visited medical facility/dispensary for treatment of respiratory issues such as cough, shortness of breath, pneumonia and other respiratory issues.

5 Gender Equality	5.4. Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Average time saving associated with faster cooking and less fuelwood collection
4 Quality Education	4.4 By 2030 substantially increase the number of youth and adults who have relevant skills including technical and vocational skills, for employment, decent jobs and entrepreneurship	Number of employees who were provided with skill development training ⁹ Number of women artisans who received training
8 Decent Work and Economic Growth	8.5. By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Total number of jobs created.
7 Affordable and clean energy	7.3 By 2030 double the global rate of improvement in energy efficiency	Total energy savings

⁹ This also includes women who were not employees but were trained in skill development programmes under the project activity.

1 No poverty	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	Average household saving i.e. decrease in expenditure on basic service such as cooking, lighting, drinking
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B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG 13: Take urgent action to combat climate change and its impacts

Parameter: Emission reductions achieved by fuelwood savings at household level

As per applied methodology 'Technologies and Practices to Displace Decentralized Thermal Energy Consumption (V3.1)', when baseline fuel and the project fuel are the same and the baseline emission factor and project emission are considered the same, the overall GHG reductions achieved by the project activity in year y are calculated as follows:

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2})) - \sum LE_{p,y}$$

Where;

$\sum_{b,p}$ Sum over all relevant (baseline b/project p) couples

$N_{p,y}$ Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y

$U_{p,y}$	Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage survey (fraction)
$P_{p,b,y}$	Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for firewood, 0.015 TJ/ton)
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass
EF_{fuel,CO_2}	CO ₂ emission factor of the fuel that is substituted or reduced (112 tCO ₂ /TJ for Firewood/Firewood waste)
$EF_{fuel,nonCO_2}$	Non-CO ₂ emission factor of the fuel that is reduced
$\Sigma LE_{p,y}$	Leakage for project scenario p in year y (tCO ₂ e/yr)

Number of project devices commissioned (N_y) will be monitored through database and monitoring surveys. The project-technology days ($N_{p,y}$) will be calculated as multiplying the number of project devices with number of operational days.

Annual usage surveys will be conducted to determine the usage rate of project devices ($U_{p,y}$). Sampling will be done for each age group of project population and weighted average will be calculated.

Quantity of fuel consumed in baseline scenario and project scenario will be determined by Kitchen Performance Tests (KPT). Specific fuel savings for a single stove ($P_{p,b,y}$) will be calculated based on the results as per the formula: $P_{p,b,y} = P_{b,y} - P_{p,y}$

Fraction of non-renewable biomass ($f_{NRB,b,y}$) is determined as using ex-ante option at the validation stage and will be fixed as 0.85 for the crediting period. The parameter has been calculated as per "Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0.

For emission factors for fossil fuels displaced by the project stoves (EF_{fuel,CO_2} and $EF_{fuel,nonCO_2}$) and net calorific value of the non-renewable woody biomass ($NCV_{b,fuel}$),

default values will be applied; i.e. 112 tCO₂/TJ for CO₂, 9.46 tCO₂e/TJ (AR5 GWP) for non-CO₂ emissions.

Leakage ($\Sigma LE_{p,y}$)

The project proponent should investigate the following potential sources of leakage:

- a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.

Reason for not considering: The project aims to displace traditional three stones stove that is already widely used in rural Malawi. It is unlikely that the project will cause the households outside the project boundaries to use higher emitting technologies or to increase their fuelwood consumption.

- b. Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.

Reason for not considering: Majority of the households rely on fuelwood in rural Malawi as per The Fifth Integrated Household Survey (Figure.1). Only 2% of the population has been reported to have access to clean cooking services in the country 2019¹⁰. No possible leakage is foreseen due to this case.

¹⁰ <https://trackingsdg7.esmap.org/country/malawi>

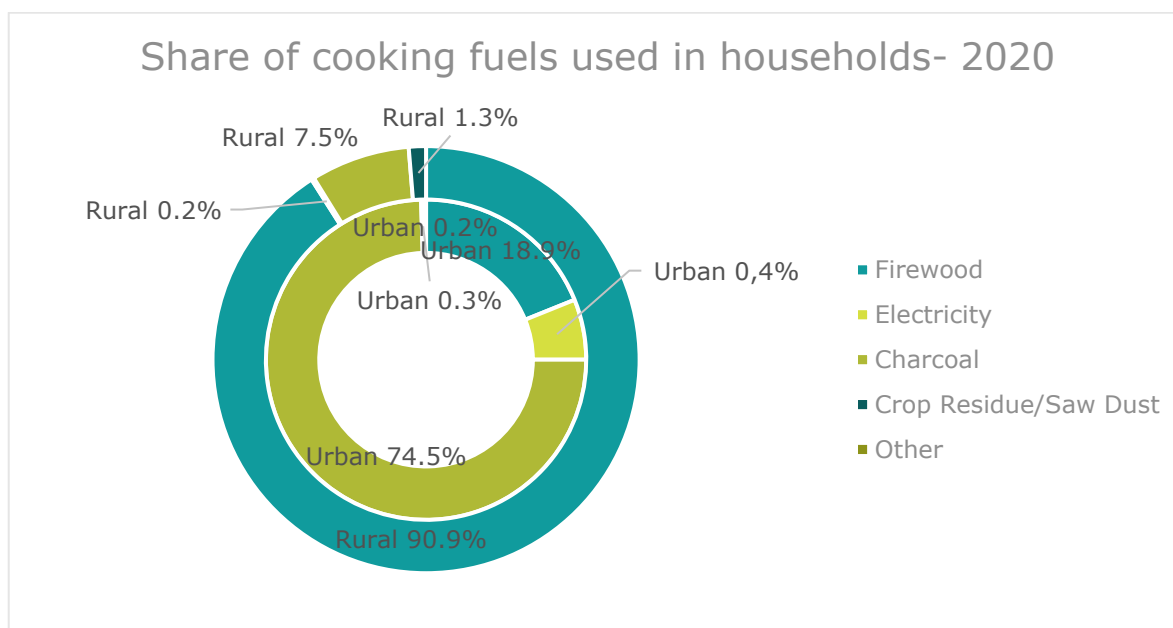


Figure 1. Share of cooking fuels used in households, Malawi, 2020.

- c. The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.

Reason for not considering: The forest cover in Malawi is depleting fast with 32% drop in less than 40 years. 99% of the Malawian population relies on biomass as cooking energy source. Illegal charcoal production due to high demand from urban areas is one of the main drivers that accelerates deforestation ¹¹.

Since firewood is collected from forest by the majority of rural population, no significant impact is expected in NRB fraction in other areas.

- d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.

Reason for not considering: Majority of the target population cook outside the house in rural Malawi according to a study carried out by 29 participants in

¹¹ https://pdf.usaid.gov/pdf_docs/PA00TFCN.pdf

2020¹²(Figure.4). Considering the climate of the country (tropical or sub-tropical depending on the altitude), space heating is not required in most of the country.

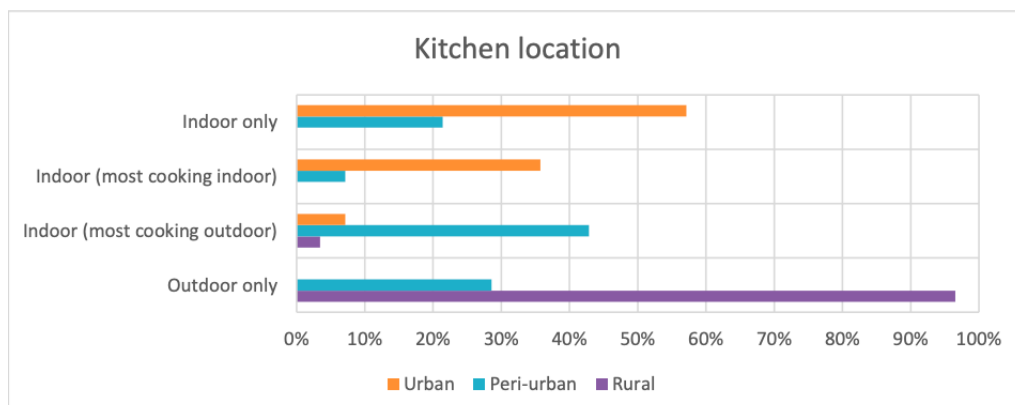


Figure 2. Places for cooking in households in Malawi, 2020

- e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.

Reason for not considering: This option is not considered as a potential source of leakage as the majority of targeted households use fuelwood and three stone fire for cooking.

SDG 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.

Parameter: Amount of firewood saved by the project stoves per year.

Project database and monitoring survey will be used to calculate total quantity of firewood saved by the project compared to baseline scenario. The following formula will be used to calculate the total amount of firewood saved:

$$Biomass_{saving} = P_{p,b,y} \times f_{NRB,b,y} \times N_{p,y}$$

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

¹² <https://meccs.org.uk/wp-content/uploads/2020/12/market-assessment-for-modern-energy-cooking-services-in-malawi-3.pdf>

Where;

- $N_{p,y}$ Project technology-days in the project database for project scenario p through year y
- $P_{p,b,y}$ Quantity of firewood that is saved in project scenario p compared to baseline scenario b during year y
- $P_{b,y}$ Quantity of firewood that is consumed in baseline scenario b during year y
- $P_{p,y}$ Quantity of firewood that is consumed in project scenario p during year y
- $f_{NRB,b,y}$ Fraction of non renewable biomass

SDG 3. Ensure healthy lives and promote well-being for all at all ages

Parameter: Proportion of households who perceive reduced indoor air pollution¹³

The beneficiaries of project stoves will be asked to evaluate any improvement in their health conditions compared to baseline situation during the household survey. By reduction of smoke due to the usage of project stoves, respiratory diseases such as itchy eyes or coughing are expected to lessen. The responses will implicitly indicate the frequency with which households were required to visit medical facilities in the pre-intervention and post-intervention scenarios.

$$MF_{reduced\ visit} = HH_{reporting\ illness,baseline} - HH_{reporting\ illness,project}$$

$$HH_{reporting\ illness,baseline} = P_{HH,illness} \times N_{p,y}$$

Where,

$MF_{reduced\ visit}$	Reduced visit to medical facility for respiration related ailments
$HH_{reporting\ illness,baseline}$	Households which report facing health issues related to indoor air pollution in baseline
$HH_{reporting\ illness,project}$	Households which report facing health issues post project implementation in a monitoring period.

¹³ This is being used as indicative of the households that would be required to visit medical facilities for treatment of their illness but owing to the population being largely rural with limited access to medical facilities and resources for treatment, cannot visit these facilities and get themselves treated.

$P_{HH, \text{illness}}$	Proportion of households that reported facing respiratory illness in the baseline survey. This value is fixed ex-ante as 66%.
$N_{p,y}$	Project technology-days in the project database for project scenario p through year y

SDG 5. Achieve gender equality and empower all women and girls

Parameter: Proportion of households who perceive reduced time for collecting wood and cooking

During the household survey, questions about the time spent on cooking and fuelwood collection will be asked to women in order to understand whether using project stoves saves time. If the response is positive, the surveyors will further explore for what kind of activities women are indulging in, in their saved time. For every household which admits to saving time, an ex-ante fixed value of reduction in time will be applied.

$$Time\ Saving_{HH,y} = (Time_{collecting\ fuel, baseline, y} - Time_{collecting\ fuel, project, y}) \times 365$$

Where,

Time Saving $_{HH, y}$	Time saved per household in a single year y.
Time $_{collecting\ fuel, baseline, y}$	Time spent by a single household per annum in collecting fuel in the baseline scenario. This value has been fixed ex-ante as 0.85 hours/day (totaling to 5.95 hours/week) derived from baseline survey
Time $_{collecting\ fuel, project, y}$	Time spent by a single household per annum in collecting fuel in the project scenario.

SDG 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

Parameter 1: Number of employees who received training during different phases of project implementation and monitoring.

Parameter 2: Number of women artisans who received training

Training records including attendance sheet, training material and photographs/video of training will be collected to monitor the number of employees/women artisans who received training.

SDG 8. Promote inclusive and sustainable economic growth, employment and decent work for all

Parameter: Number of temporary and permanent jobs created

Records of number and type of jobs created with employment status, gender representation and duration.

SDG 1. End Poverty in all its form everywhere

Parameter: Average household saving on fuelwood i.e. decrease in expenditure on purchase of fuel wood on account of its reduced consumption.

Average annual savings per household which was purchasing fuelwood in the pre-intervention scenario.

From the baseline study it was determined that nearly 31% of the end users were purchasing firewood. The same study also pointed out that on an average these households purchased on an average 12.50% of all the firewood which was consumed for meeting cooking needs. The price of one ton of firewood is 81395 Malawian Kwacha as per baseline survey. Thus, annual savings associated with reduction in purchase of firewood per household is:

$$Saving_{HH,y} = (P_{p,b,y} \times 0.125) \times 81395$$

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

P_{p,b,y}	Quantity of firewood that is saved in project scenario p compared to baseline scenario b during year y
P_{b,y}	Quantity of firewood that is consumed in baseline scenario b during year y
P_{p,y}	Quantity of firewood that is consumed in project scenario p during year y
81395	Average price of firewood in the region of project implementation as determined from baseline study. (Kwacha/ton)
0.125	Percentage of fuelwood consumption which was purchased in baseline

SDG 7. Ensure access to affordable, reliable, sustainable and modern energy for all

Parameter: Total energy saved

Total savings in terms of reduced thermal energy demand.

$$Energy\ Saving = N_{p,y} \times P_{p,b,y} \times NCV_{b,fuel}$$

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

Where,

$N_{p,y}$	Project technology-days in the project database for project scenario p through year y
$P_{p,b,y}$	Quantity of firewood that is saved in project scenario p compared to baseline scenario b during year y
$P_{b,y}$	Quantity of firewood that is consumed in baseline scenario b during year y
$P_{p,y}$	Quantity of firewood that is consumed in project scenario p during year y
$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for firewood, 0.015 TJ/ton)

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	$EF_{fuel,CO2}$
Unit	tCO ₂ /TJ
Description	CO ₂ emission factor arising from use of fuels in baseline and project scenario
Source of data	Technologies and Practices to Displace Decentralized Thermal Energy Consumption" version 3.1 August 2017
Value(s) applied	112

Choice of data or Measurement methods and procedures	Default value
Purpose of data	Calculation of emission reductions
Additional comment	

Data/parameter	$EF_{fuel, non-CO2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in baseline and project scenario
Source of data	TPDDTEC methodology version 4.0 default
Value(s) applied	9.46
Choice of data or Measurement methods and procedures	Methodology Default
Purpose of data	Calculation of emission reductions
Additional comment	

Data/parameter	$NCV_{b, fuel}$
Unit	TJ/tonne
Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	IPCC default value
Value(s) applied	0.015
Choice of data or Measurement methods and procedures	Default value
Purpose of data	Calculation of emission reductions

Additional comment	N/A
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SDG 3

Data/parameter	Proportion of households suffering from indoor air pollution in baseline
Unit	Fraction
Description	Members in the households who are experiencing respiratory health problems from inhaling smoke from baseline cookstoves
Source of data	Baseline survey
Value(s) applied	66%
Choice of data or Measurement methods and procedures	Survey carried out in control population
Purpose of data	Assessment of contribution to SDG 3
Additional comment	N/A

SDG 5

Data/parameter	Average fuel collection time in baseline
Unit	hours/day
Description	The average time spent in collecting fuel wood per day in baseline
Source of data	Baseline survey results
Value(s) applied	0.85
Choice of data or Measurement methods and procedures	Survey carried out in control population
Purpose of data	Assessment of contribution to SDG 5
Additional comment	N/A

Data/parameter	Average time saved in project households
Unit	hours/week
Description	The average time saved in collecting fuel wood per week in project households owing to reduced wood consumption.
Source of data	Berkeley Air Report- Effects on gender-related outcomes after the introduction of improved cookstoves in rural Zambia
Value(s) applied	4
Choice of data or Measurement methods and procedures	Study on same model of cookstove in Zambia showed saving of 4 hours/week of drudgery in wood collection. As the time spent in collecting wood is proportional to the amount consumed by the stove for cooking meals, hence a similar saving can be expected for the project activity as well.
Purpose of data	Assessment of contribution to SDG 5
Additional comment	N/A

SDG 1

Data/parameter	Percent households purchasing fuelwood in baseline
Unit	Per cent
Description	Percentage of end users who are purchasing fuelwood in the baseline
Source of data	Baseline survey
Value(s) applied	31
Choice of data or Measurement methods and procedures	Survey carried out in control population
Purpose of data	Assessment of contribution to SDG 1
Additional comment	N/A

Data/parameter	Percent of total consumption which is purchased
Unit	Per cent
Description	Average percentage of total fuel consumption that a household purchases.
Source of data	Baseline Survey
Value(s) applied	12.5
Choice of data or Measurement methods and procedures	Survey carried out in control population
Purpose of data	Assessment of contribution to SDG 1
Additional comment	N/A

Data/parameter	Cost of firewood
Unit	M Kwacha per ton
Description	Average price of firewood in the baseline
Source of data	Baseline Survey
Value(s) applied	81395
Choice of data or Measurement methods and procedures	Survey carried out in control population
Purpose of data	Assessment of contribution to SDG 1
Additional comment	N/A

B.6.3. Ex ante estimation of SDG Impact

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SDG 13: Take urgent action to combat climate change and its impacts

Parameter: Emission reductions achieved by fuelwood savings at household level

Ex-ante calculation for a project stove by using the parameter values fixed ex-ante and the preliminary values defined for monitored parameters in B.7.1 is demonstrated as follows:

Parameter	Unit	Data Source	Value
$N_{p,y}$	Number	Project technology-days in the project database for project scenario p through year y	1
$U_{p,y}$	Fraction	Assumed	0.90
$P_{p,b,y}$	t/hh/d	Assumed Field performance test results (KPT)	$P_{p,b,y} = P_{b,y} - P_{p,y}$ $P_{p,b,y} = 3.26 \text{ t/hh/y} - 0.94 \text{ t/hh/y} = 2.31 \text{ t/hh/y}$
$NCV_{b,fuel}$	TJ/t	Default value for firewood	0.015
$f_{NRB,b,y}$	Fraction	NRB Report	0.85
$EF_{fuel,CO2}$	tCO ₂ /TJ	Default value for firewood	112
$EF_{fuel,nonCO2}$	tCO ₂ /TJ	Calculated from IPCC	9.46
$\Sigma LE_{p,y}$	tCO ₂	Assumed to be zero as per the methodology	0

$$ER_y = \Sigma_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2}) - \Sigma LE_{p,y})$$

Emission reductions achieved per project stove is calculated as follows

$$ER_y = (1 \text{ year}) \times 0.90 \times 2.31 \frac{t}{hh} \times 0.015 \frac{TJ}{Ton} \times (0.85 \times 112 + 9.46 \text{ t/TJ}) - 0 = 3.23 \text{ ton/hh}$$

SDG 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

Parameter: Amount of firewood saved by the project stoves per year

Amount of firewood saved will be calculated by the formula:

$$Biomass_{saving} = P_{p,b,y} \times f_{NRB,b,y} \times N_{p,y}$$

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

P_{b,y}	Baseline fuel consumption (Tons/HH/Yr)	3.26
P_{p,y}	Project fuel consumption (Tons/HH/Yr)	0.94
P_{p,b,y}	Calculated (Tons/HH/Yr)	P_{p,b,y} = P_{b,y}-P_{p,y}
		2.31
<i>f_{NRB,b,y}</i>	Third party report	85%
Biomass saving	Calculated (Tons/hh/year)	2.01

SDG 3: Ensure Healthy lives and promote well-being for all at all ages

Parameter: Proportion of households who perceive reduced indoor air pollution

$$MF_{reduced\ visit} = HH_{reporting\ illness,baseline} - HH_{reporting\ illness,project}$$

$$HH_{reporting\ illness,baseline} = P_{HH,illness} \times N_{p,y}$$

P _{HH, illness}	Proportion of households experiencing respiratory illness associated with indoor air pollution	66%
N _{p,y}	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y	1
HH _{reporting illness, project}	Households reporting respiratory illness in post project scenario	Assumed 0, actual value will depend on project survey

$$HH_{reporting\ illness,baseline} = 0.66 \times 50,000$$

$$MF_{reduced\ visit} = 33000 - 0$$

$$MF_{reduced\ visit} = 33000$$

SDG 5: Achieve Gender Equality and empower all women and girls

Parameter: Proportion of households who perceive reduced time for collecting wood and cooking

$$Time\ Saving_{HH,y} = (Time_{collecting\ fuel,baseline,y} - Time_{collecting\ fuel,project,y}) \times 365$$

Time _{collecting fuel, baseline, y}	Time spent in collecting fuel in baseline. Fixed ex-ante.	0.85 hours/hh/day
Time _{collecting fuel, project, y}	Time spent in collecting fuel in project scenario. Calculated based on average time saving of 4 hours/week	(0.9-0.57) = 0.28 hours/hh/day

Total operational days	Total number of days when the stove was operational	365 (as the stoves are installed in households)
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$$Time\ Saving_{HH,y} = (0.85 - 0.28) \times 365$$

= 209 hours/ year for all those households which report reduction in fuel collection time post project intervention.

For **SDG 8:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all

Parameter: Employment by type, gender and duration. Sample of employment contracts along with records of individuals employed for various durations for specific activities during the monitoring period.

SDG 1: End poverty in all its form everywhere

Parameter: Average household saving on fuelwood i.e. decrease in expenditure on purchase of fuel wood on account of its reduced consumption.

$$Saving_{HH,y} = (P_{p,b,y} \times 12.5 \times f_{NRB}) \times 81395$$

$$P_{p,b,y} = (P_{b,y} - P_{p,y})$$

$P_{b,y}$	Baseline fuel consumption (Tons/HH/Yr)	3.2558
$P_{p,y}$	Project fuel consumption (Tons/HH/Yr)	0.9437
$P_{p,b,y}$	Calculated (Tons/HH/Yr)	$P_{p,b,y} = P_{b,y} - P_{p,y}$ 2.3121
Percent of total consumption purchased	Percentage of fuelwood consumption that was purchased in the baseline	12.5%
Percent households which purchased fuelwood	Percentage of end users who were purchasing fuelwood in baseline	31%
$f_{NRB,b,y}$	Third party report	85%
Unit cost of firewood	Baseline survey (M Kwacha /ton firewood)	81395

$$Saving_{HH,y} = (2.3121 \times 0.125 \times 0.85) \times 81395$$

=19,977 M Kwacha/year for each household which was purchasing fuelwood in the baseline

SDG 7. Ensure access to affordable, reliable, sustainable and modern energy for all

Parameter: Total energy saved

Total savings in terms of reduced thermal energy demand.

$$Energy\ Saving = N_{p,y} \times P_{p,b,y} \times NCV_{b,fuel}$$

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

$P_{b,y}$	Baseline fuel consumption (Tons/HH/Yr)	3.2558
$P_{p,y}$	Project fuel consumption (Tons/HH/Yr)	0.9437
$P_{p,b,y}$	Calculated (Tons/HH/Yr)	$P_{p,b,y} = P_{b,y} - P_{p,y}$
		2.3121
$N_{p,y}$	Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y	1
$NCV_{b,fuel}$	Net calorific value of biomass (TJ/Ton)	0.015

For SDG 4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all

Parameter: Training records.

Training records with training material and attendance sheets.

B.6.4. Summary of ex ante estimates of each SDG outcome

SDG 13 Parameter: Climate change mitigation (Amount of GHGs emissions avoided or sequestered)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	115,003	33,335	81,668
2024	230,007	66,669	163,338
2025	230,007	66,669	163,338
2026	230,007	66,669	163,338
2027	230,007	66,669	163,338
Total	1,035,031	300,011	735,020

Total number of crediting years 5

Annual average over the crediting period	207,006	60,002	147,004
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SDG 15 Parameter: Total non-renewable wood fuel saved(tonnes/y)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	69,185	20,054	49,131
2024	138,371	40,108	98,263
2025	138,371	40,108	98,263
2026	138,371	40,108	98,263
2027	138,371	40,108	98,263
Total	622,669	180,486	442,183

Total number of crediting years 5

Annual average over the crediting period	124,533	36,098	88,435
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SDG 3 Parameter: Number of households visited medical facilities/dispensary for treatment of respiratory issues etc. such as cough, shortness in breath, pneumonia and other respiratory issues (Number).

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	16,500	0	16,500
2024	33,000	0	33,000
2025	33,000	0	33,000
2026	33,000	0	33,000
2027	33,000	0	33,000
Total	148,500	0	148,500

Total number of crediting years 5

Annual average over the crediting period	29,700	0	29,700
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SDG 5 Parameter: Average time saving associated with faster cooking and less fuelwood collection (hr/hh)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	310	102	209
2024	310	102	209
2025	310	102	209
2026	310	102	209
2027	310	102	209
Total	1551	508	1,043
Total number of crediting years 5			
Annual average over the crediting period 310 102 209			

SDG 4 Parameter: Number of employees provided skill development training.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	0	20	20
2024	0	0	0
2025	0	0	0
2026	0	0	0
2027	0	0	0
Total	0	20	20
Total number of crediting years 5			
Annual average over the crediting period 4			

SDG 8 Parameter: Total number of people employed (full-time & part-time)

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	0	25	25
2024	0	25	25
2025	0	0	0
2026	0	0	0
2027	0	0	0
Total	0	50	50

Total number of crediting years 5

Annual average over the crediting period 10

SDG 7 Parameter: Total energy savings

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	1,220	354	866
2024	2,441	708	1,733
2025	2,441	708	1,733
2026	2,441	708	1,733
2027	2,441	708	1,733
Total	10,984	3,186	7,798

Total number of crediting years 5

Annual average over the crediting period 2,285 638 1,558

SDG 1 Parameter: Average household saving i.e. decrease in expenditure on basic service such as cooking, lighting, drinking.

YEAR	BASELINE ESTIMATE	PROJECT ESTIMATE	NET BENEFIT
2023	28,156	8,161	19,995

2024	28, 156	8,161	19,995
2025	28, 156	8,161	19,995
2026	28, 156	8,161	19,995
2027	28, 156	8,161	19,995
Total	140,780	40,805	99,975
Total number of crediting years 5			
Annual average over the crediting period	28,156	8,161	19,995

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	$N_{p,y}$
Unit	Project technologies credited (unit)
Description	Technologies in the project database for project scenario p through year y
Source of data	Project Database Records
Value(s) applied	50,000 (assumed)
Measurement methods and procedures	Detailed sales information will be collected for each customer, either using electronic or paper-based means. On electronic means, the data will be collected by SMS (i.e. mobile phone 'short message service' text) or Information and Communication Technologies ('ICT' – such as PDAs). Information that is entered into the

	database includes the name of the customer, address/ description of location, contact telephone number(s), unique serial number of the stove, retailer ID, and date of purchase. Sales information submitted via SMS/ ICT will automatically enter the database. Written registration cards will be entered manually into the same database.
Monitoring frequency	Continuously
QA/QC procedures	Project distribution staff will spot-check end-users to verify that information submitted was factual.
Purpose of data	Cross checking Calculation of emission reductions
Additional comment	N/A

SDG 13

Data / Parameter	$U_{p,y}$
Unit	Fraction
Description	Cumulative usage rate for technologies in project scenario p during year y, based on cumulative installation rate and drop-off rate.
Source of data	Usage Survey
Value(s) applied	90% (assumed)
Measurement methods and procedures	The proportion of sampled ICS found to be in operation during each monitoring period will be applied to the total number of stoves accounted for emission reductions. Minimum 100 samples will be surveyed. Separate samples of at least 30 shall be taken for each age group of project population and weighted average factor for overall project population shall be used.
Monitoring frequency	Annually

QA/QC procedures	The unique reference number of each stove shall be logged in the monitoring database showing the total number of stoves. Data from the sampling plan will be collected in each monitoring period by trained project staff and applied in the emissions reduction calculations. Internal cross-checks by the project developer or project implementer will be undertaken as QC
Purpose of data	Calculation of emission reductions
Additional comment	N/A
Data / Parameter	fNRB
Unit	Fraction
Description	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass
Source of data	Calculated
Value(s) applied	0.85
Measurement methods and procedures	Calculated as per "Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0
Monitoring frequency	Fixed by baseline study for a given crediting period
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data / Parameter	$P_{b,y}$
Unit	kg/stove/day
Description	Quantity of firewood that is consumed in baseline scenario b during year y
Source of data	Baseline FT, Baseline FT updates and any applicable adjustment factors
Value(s) applied	3.26 t/hh/y (assumed)

Measurement methods and procedures	Determined by the baseline KPT carried out sampling the households without the project stove from targeted population. For paired sampling, the sample size of stove users shall be chosen for a 90/30 precision (90% confidence interval+ and 30% margin of error). In cases where the result indicates that 90/30 precision is not achieved, the lower limit of 30% confidence interval of the parameter value will be chosen as an alternative to repeating the survey efforts to achieve the 90/30 precision.
Monitoring frequency	Whenever required to update the baseline.

QA/QC procedures	Transparent data analysis and reporting, the latest protocol for Kitchen Performance Test ¹⁴ will be applied.
Purpose of data	Calculation of emission reductions
Additional comment	The baseline is fixed ex-ante during the crediting period as the project is targeting non-industrial application (TPDDTEC, Page 8). A single baseline fuel consumption parameter is weighted to be representative of baseline technologies being compared for project crediting.

Data / Parameter	$P_{p,y}$
Unit	kg/stove/day
Description	Quantity of firewood that is consumed in project scenario p during year y
Source of data	Project FT, and any applicable adjustment factor
Value(s) applied	0.94 t/hh/y
Measurement methods and procedures	Ex ante: Calculated based on 34.5% thermal efficiency reported Ex Post: The latest protocol for Kitchen Performance Test¹⁵ will be applied. The sample size of stove users shall be chosen for a 90/30 precision (90% confidence interval+ and 30% margin of error). In cases where the result indicates that 90/30 precision is not achieved, the lower limit of 30% confidence interval of the parameter value will be chosen as an alternative to repeating the survey efforts to achieve the 90/30 precision.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Transparent data analysis and reporting

¹⁴ <https://www.cleancookingalliance.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

¹⁵ <https://www.cleancookingalliance.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

Purpose of data	Calculation of emission reductions
Additional comment	A single project fuel consumption parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario

Data / Parameter	$LE_{p,y}$
Unit	tCO ₂ e
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring surveys
Value(s) applied	0
Measurement methods and procedures	Measured through baseline surveys and monitoring surveys
Monitoring frequency	Biennially
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of leakage emissions
Additional comment	

Data / Parameter	η_{new}
Unit	Fraction
Description	Efficiency of the ICS installed as a part of the project activity.
Source of data	Water Boiling Tests following the latest WBT protocol
Value(s) applied	-
Measurement methods and procedures	As per Annex 8 of applied methodology TPDDTEC(V3.1): - The minimum sample size of each age group shall comply with the 90/10 rule. - The tests will be carried out in the last 3 months of the monitoring period.

	Samples will be selected as per the Guidelines for sampling and surveys for CDM project activities and programme of activities.
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of project fuel consumption
Additional comment	Only applicable when the ageing test approach (Annex 8 of applied methodology TPDDTEC, v3.1) is preferred.

Data / Parameter	μ_y
Unit	Fraction
Description	Adjustment factor continued use of pre-project devices during the year y
Source of data	Monitoring surveys
Value(s) applied	-
Measurement methods and procedures	Any of the following approach with justification on how the chosen option fits within project circumstances - Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity - Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidences to determine the frequency of usage of both the project devices and baseline devices. - Monitoring surveys to capture the number of meals cooked
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions

Additional comment	Only applicable when the ageing test approach (Annex 8 of TPDDTEC) is preferred.
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SDG 8

Data / Parameter	Number of jobs created by the project activity
Unit	Number
Description	Temporary and permanent jobs created for building stove, maintenance, repair, sale and distribution, monitoring etc.
Source of data	Employment records and contracts
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	The employment records by gender of the worker and contracts will be used to monitor the parameter.
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 8
Additional comment	N/A

SDG 5

Data / Parameter	Proportion of households who report reduced time in collecting wood and cooking
Unit	Fraction
Description	Households which report reduction in fuel wood collection time owing to reduced consumption in project stoves.
Source of data	Project Survey
Value(s) applied	N/A
Measurement methods and procedures	Survey Questionnaire
Monitoring frequency	Annually/biennially

QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 5
Additional comment	N/A

SDG 4

Data / Parameter	Number of employees who received skill development training
Unit	Number
Description	Count of employees/women who attended training sessions for skill development
Source of data	Training records
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	N/A
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 4
Additional comment	This category also includes female end-users who received skill development training such as knitting/stitching etc, which is helping them generate income

SDG 3

Data / Parameter	Proportion of households suffering from indoor air pollution post intervention
Unit	Fraction
Description	Members in the households experiencing health problems from inhaling smoke in project scenario requiring visit medical facility
Source of data	Project survey

Value(s) applied	0% (assumed)
Measurement methods and procedures	Survey questionnaire
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 3
Additional comment	N/A

SDG 1

Data / Parameter	Total quantity of fuel wood purchased post intervention
Unit	Tons
Description	Total quantity of fuelwood purchased in project scenario
Source of data	Project Survey
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	Survey Questionnaire
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 1
Additional comment	N/A

B.7.2. Sampling plan

>>

The sampling frame is the project boundaries, including the houses where the stoves have been installed. The target population are the households in rural Malawi. The end users who received the Project stoves will be recorded to the sales database.

To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined

sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size.

Baseline Survey:

For baseline survey, samples will be randomly selected among households without the project technology. The minimum sample size required by the methodology is 100 for population over 1000.

Kitchen Performance Test:

The sampling method for $P_{b,y}$ and $P_{p,y}$ is Paired Sampling or sampling the same users before and after beginning use of the project stove. Sample size must be sufficient to ensure the confidence/precision level of 90/30 for quantity of fuelwood consumed in baseline and project scenario. The minimum sample size recommended by the methodology is 20 for KPTs.

The baseline KPT will be performed with 30 samples that will be randomly selected among the over 100 end users in baseline survey. The KPT will be performed on houses with similar socioeconomic characteristics as the target population. After baseline KPT is finalized, the households will be provided with the project stoves for the purpose of the test. The project KPT will be run after allowing sufficient time to the subjects to get used to the project stoves.

Subsequent sampling for updating $P_{p,y}$ will be done by stratified sampling method per stove ages. The registered stoves will be grouped by age and location from the project database. The sample size updating KPT will comply with 90/30 confidence/precision required by the methodology.

Water Boiling Test:

PP may opt to follow aging test approach for project fuel updates as per Annex.8 of the methodology. In that case, annual WBTs will be conducted with a representative sample of each age group instead of biennial KPTs. The first WBT will be carried out along with the project KPT prior to the first issuance in the field or laboratory, following the latest version of WBT protocol by an independent expert or entity. Subsequent WBTs will be carried out for monitoring the degradation of efficiency of ICS on annual basis. This shall be done during the last 3 months of monitoring provided the chosen period is representative of annual conditions.

Sample selection shall follow the recent Guidelines for Sampling and Surveys for CDM Project Activities and Programme Activities. The sample size shall comply with 90/10 confidence/precision level for each age group.

Usage Rate:

For $U_{p,y,r}$, simple random sampling method per stove ages will be followed. Sample size for usage rate is minimum 100 for group sizes larger than 1,000.

Usage Rate Monitoring guidelines identifies the requirements as follows:

Level	Applicability	Claimable usage rate	Requirements
Mandatory	Mandatory	Upto maximum 75%	- Define use and non-use - In person household usage survey - Verification of accuracy of results
Good Practice	Optional	Upto maximum 90%	- Field team training and supervision - End -user training and follow ups - Awareness campaign
Best Practice	Optional	Above 90%	Continuous use monitoring

Monitoring Surveys:

The monitoring survey investigates changes over time in the project scenario by surveying end-users with project technology on an annual basis. Annual trends in end user characteristics such as stove use, fuel consumption and seasonal variations are predicted during the survey. Whether the baseline technology is still in use will be checked and recorded as well.

Leakage shall be assessed biennially as per methodology. It will be done along with the monitoring survey.

B.7.3. Other elements of monitoring plan

>>

The Monitoring Plan applied involves a number of key elements that ensure high-quality, unbiased and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Emission Reductions (VERs) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

Below is the description of the steps in monitoring plan.

1. **Registration of stove:** Project Implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a hard copy form or ICTs by project implementer's staff, retailers, end-users or partner organization's staff, or through the use of ICTs or SMS. Project Implementers' staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed. Following information will be recorded:
 - Serial Number
 - Contact details of the user
 - Date of installation
 - GPS Coordinates

- Number of stoves installed

Every beneficiary of an ICS will sign a user agreement (paper and/or electronic version) with Project Developer. A unique serial number will be allocated to each stove and the number is indicated on the user agreement.

2. Data logged into database: Project Implementer's trained staff will input the data in the database either manually (if data collected using hard copy form) or this will be automatically input if data was collected using ICTs or SMS. Project Implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated and errors corrected or excluded from the database if it is a true duplicate entry.

3. Spot- checking (continuously): Project Implementer field staff will randomly select households included in the database and visit them to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (eg. change in the address of a user) will be updated on the database, and in the case ICS are found to be no longer in use, they will be clearly marked as such and excluded from emission reductions calculations.

4. Monitoring: Project Implementer will follow the requirements as per methodology requirements to collect the necessary information for a monitoring report.

5. Preparation of monitoring report: Project Implementers or Project Developer will prepare the final monitoring report to be provided to the verifier VVB for verification of emission reductions.

Project Developer will coordinate and manage Project Implementer and assist them in implementing each element of the monitoring plan.

Biennial Project FT Update:

The PFT update is an extension of the project KPT and done biennially. The test provides a fuel consumption assessment representative of project technologies currently in use and predicts any changes in the project scenario over time as project stove age and new customers are added.

Annual Monitoring and Usage Surveys:

According to the methodology, end users can be surveyed at any time(s) throughout the year with care taken to collect information pertaining to seasonal variations in stove and fuel use patterns. Monitoring surveys can be conducted with usage survey participants that are currently using the project stove. Visual inspection of the premises to see if the project stoves are operational and in use and interview with end user will be done during the survey.

In case WBTs opted:

If aging test approach is followed, Project proponent will have to monitor parallel use of baseline stove in the project scenario by any of the following options:

- Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity
- Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidence to determine the frequency of usage of both the project devices and baseline devices.
- Monitoring surveys to capture the number of meals cooked

Baseline FT Update:

If baseline is required to be updated any time during the crediting period, baseline performance field tests will be carried out as per described above.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of VPA

>>

The installation of the stoves is estimated to start on 01/07/2023

C.1.2. Expected operational lifetime of VPA

>>

15 years

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

01/07/2023

C.2.2. Total length of crediting period

>>

5 years renewable twice, 15 years of total crediting period.

01/07/2023 to 30/06/2028 (Expected crediting period)

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles	Mitigation Measures added to the Monitoring Plan
	Necessary precautions shall be undertaken by the Project Implementers
Principle 3	during the stove installation. First aid kits shall be available in workshops. Trainings will be provided for all workers
Principle 9.7	The users will be trained on how to harvest trees and which parts could be fuelwood.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project integrates gender sensitive design and gender aspect in all phases of the project. Women and youth will benefit mostly from the efficient cookstoves installed in households and therefore their feedback will be considered during the design and implementation of the project. Moreover, the project plans to organize trainings for women and community artisans on making bricks and building their own stoves. This will increase the participation of women and empower them in the society.
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Question 2 - Explain how the project aligns with existing country policies, strategies and best practices

National Gender Policy¹⁶ for Malawi was adopted in January 2015. Women are hardest hit by effects of environmental mismanagement because of the gender roles they play in the resource utilization as stated in the policy paper. However, women have limited participation and representation as low as 23% in decision making process at all levels. Under representation of vulnerable groups in decision making forums on the utilization of forests, land and water and inadequate information and education on sustainable environmental management are among the gender related issues within natural resources, environment and climate change management.

Traditionally, women and children especially girls collect firewood that is the major source of energy in most households especially in rural Malawi. They have to walk long distances as the source of wood fuel is usually far from settlements due to deforestation and other reasons. This makes children too tired to concentrate at school as girls has a higher drop-out rate then boys in Malawi. Gender based violence against women and girls are on the increase in the country. They have to face health

¹⁶ <https://cepa.rmportal.net/Library/government-publications/National%20Gender%20Policy%202015.pdf>

	problems such as persistent back aches, ill health among pregnant women and snake bites as well. Furthermore, they are exposed to open flames and smoke during cooking which result in respiratory problems. The project addresses gender issues related with energy by installing improved cookstove technologies for households. Women and girls, being the main beneficiaries of the project are actively involved in all phases of the project including the design of cookstoves and participating stakeholder consultation. The project will further provide training to women artisans to make bricks and build their own stoves and will record the number of women employed. This empowers women by generating income and raise their status in the family.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Gender issues raised by Gold Standard Safeguarding principles are all addressed under the Safeguarding Principles Assessment in Appendix.1. No risks have been foreseen by the stakeholders and project developer.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	As per Stakeholder Consultation Requirements, the project developer invites all stakeholders to give feedback on the design and the implementation of the project. During the consultation

process, particular importance is given to the equal participation of women. A group of beneficiaries who will be using the technology have been invited to the live meeting and their trip was arranged by the project developer. Moreover, gender related organizations who actively work in rural areas have been invited to the meeting.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

The stakeholders supported the project and confirmed the contributions to sustainable development. Their general concern was how implementation will be carried out as there are cookstoves already implemented and some of them are said to be unsuccessful. C-Quest Capital have sufficient experience on implementation of cookstove projects in Sub Saharan Africa and evaluates all risks before kicking off any projects. The potential partners will be evaluated through standard procedures and selected on due diligence.

E.2. Final continuous input / grievance mechanism

METHOD	INCLUDE ALL DETAILS OF CHOSEN METHOD (S) SO THAT THEY MAY BE UNDERSTOOD AND, WHERE RELEVANT, USED BY READERS.
Continuous Input / Grievance Expression Process Book (mandatory)	The process books will be placed in the C-Quest Capital's local partners' offices once the contracts are signed.
GS Contact (mandatory)	help@goldstandard.org
Other	Tel: +1 202 416 2400 Tel (Malawi): +265 99 986 06 25 OR +265 999 10 73 79 Cqc-operations@cquestcapital.com

SECTION F. Eligibility and inclusion criteria for VPAs inclusion

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The below table shall be completed for all VPAs.

The CME shall provide clear description on how eligibility criteria set at real case VPAs are complied with for each real case and regular VPAs submitted for inclusion.

The CME shall not change the eligibility criteria and required condition set at real case VPAs. At the time of inclusion of regular VPAs, the CME shall only describe how the regular VPAs comply with the eligibility criterion.

NO.	ELIGIBILITY CRITERION	DESCRIPTION/ REQUIRED CONDITION	DESCRIPTION OF THE VPA IN RELATION TO THE CRITERIA, MEANS OF VERIFICATION/SUPPORTING EVIDENCE FOR INCLUSION
1	Geographic boundary	Each VPA shall be located within the boundaries of the PoA.	The VPA has been implemented in Malawi which is included in the geographic boundary of The PoA. CME declaration cross verified with stove installation/registration database.
2	Double Counting	A unique numbering system will be adopted for each VPA. All ICS installed shall receive a unique number identifying to which VPA it belongs.	Each ICS included in the VPA will have a unique combination of customer name and geographical location linked with a unique serial number. Evidence of unique numbering system, database of stoves implemented in the region in different GHG programmes to be cross checked with the project stove database.

3	Exclusiveness of VPA	VPA shall not be previously: 1. Registered as a project activity with other offset schemes 2. Included as a VPA in any other registered PoA, or deregistered as a VPA of a PoA.	The current VPA is a new activity implemented under the PoA. Cross verification inclusion of the VPA under other programmes.
4	Specification of Technology/ Measure	VPAs under the PoA shall install in-situ ICS using firewood. The capacity is limited by 150kW as per the applied TPDDTEC methodology.	The stoves being implemented under the CPA are fixed TLC rocket stoves. Each stove is assumed to have a thermal capacity of 1.61 kW. The actual capacity calculation shall be submitted to verifying VVB.
5	Start date	The start date of any proposed VPA will be the date of installation of first device under it.	The date on which first ICS installed will be recorded. Sales invoice or end-user agreement will be provided as a proof. The start date of the VPA is 01/07/2023 which is the expected date of installation of first stove included under it.
6	Applicability of methodologies	VPAs shall apply methodology- Technologies and Practices to Displace Decentralized Thermal Energy Consumption (v 3.1).	Please refer to section B.2 of VPA-DD for confirmation.
7	Additionality	The additionality PoA is demonstrated as per GS4GG Community services activity requirements, Version 1.2	Confirmation by the CME that the project is located in Malawi to be provided to the VVB. Total thermal energy savings per unit is 0.009634 GWh as per assumed KPT and WBT results.
8	Public funding	Affirmation that there is no diversion of Official Development Assistance (ODA).	Declaration of CME

9	Target group & Distribution Mechanism	Households will be the target group for VPAs. ICSs will be installed by local VPA Implementers on a non-commercial basis.	CME declaration or user registration database.
10	Sampling requirements	All requirements as mentioned in TPDDEC, version 3.1 or the Standard: Sampling and surveys for CDM project activities and programme of activities are applicable to VPAs.	Please see Section B.7.2 Sampling Plan for details. VPA Implementers will follow the management system described at the PoA-DD.
11	Compliance of the technology implemented	The capacity of each ICS will comply with the requirements of TPDDTEC methodology.	VPA applies GS methodology- Technologies and Practices to Displace Decentralized Energy Consumption, (v3.1) Please refer to Section B.2 of the VPA-DD for demonstration of its compliance with TPDDTEC requirements.
12	SDG Outcomes	Each VPA shall conduct SDG outcomes assessment and comply with the SDG targets identified in the PoA-DD.	VPA-DD includes SDG outcomes assessment as defined in the PoA. Please refer to Section B.6 of the VPA-DD. SDG Assessment calculation sheet to be provided to the VVB at the time of verification.
13	Stakeholder Consultation	Each VPA will conduct a local stakeholders' consultation.	Local Stakeholder Consultation Meeting is conducted on July, 29 th 2021. Please see Stakeholder Consultation Report for details.

14	VER Ownership	End users receiving ICSs under the specific VPA contractually cede their rights to claim and own emission reductions to the CME of the PoA.	A sample carbon waiver document has been provided to VVB.
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APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	No	1. The Republic of Malawi is a member of the United Nations and the African Union. It has ratified many UN Human Rights Conventions and thus has made binding international commitments to adhere to the standards laid down in these universal human rights	N/A

		documents. ¹⁷ The project will be implemented under the national laws and will not lead to violations of human rights in any kind.	
		There is no limitation to the participation to the project.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of non-discrimination, equal	No	1. The project aims to decrease the burden on women in the most vulnerable communities by reducing fuel wood consumption. The time spending for fuel wood	N/A

¹⁷ <http://www.claiminghumanrights.org/malawi.html?L=0>

treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		collection and cooking will reduce. The women will have more time for other activities. The risk of being exposed to gender-based violence will decrease as well. 2. Both women and men will have equal access to the project stoves and other benefits. The project will provide job opportunities and aims to provide trainings to women artisans for building project stoves. Principles of equal treatment, equal pay for equal work will be strictly followed. 3. The Project respects the country's gender	
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		policy. The project addresses gender issues related with energy by installing improved cookstove technologies for households. Women and girls, being the main beneficiaries of the project are actively involved in all phases of the project including the design of cookstoves and participating stakeholder consultation. 4. N/A	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks	Potentially	The project will follow the requirements in Occupational Safety, Health and Welfare	Necessary precautions shall be undertaken by the Project Implementers during the

and shall not adversely affect the health of the workers and the community		Act 1997 ¹⁸ . The Project Implementers will provide safe workplace, machinery and equipment for the artisans.	stove installation. First aid kits shall be in available in workshops. Trainings will be provided for all workers.
Principle 4.1 Sites of Cultural and Historical Heritage			
Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project does not involve and is not complicit in the alteration, damage or removal of any critical cultural heritage.	
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not require or cause the involuntary relocation of people. The project stoves will be installed in houses.	
>>			

¹⁸ http://www.oit.org/dyn/natlex/natlex4.detail?p_lang=en&p_isn=57356&p_country=MWI&p_count=105&p_classification=14&p_classcount=6

Principle 4.3 Land Tenure and Other Rights			
a. Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No	The project does not require any change to land tenure arrangements and/or other rights such as resource access rights, community-based property rights and customary rights.	
>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project does not require any land for implementation.	NA
>>			
Principle 5. Corruption			

1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	No	Corrupt Practices Act 1995¹⁹ of Malawi establishes an Anti-Corruption Bureau with the function of investigating corruption in public and private bodies, sets forth offences against the Act. The Bureau can investigate public officials who appear to be living beyond their means. The project participants will not be involved, complicit or contribute towards corruption according to the Act.	NA
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national	No	1. Malawi ratified ILO N°87 Freedom of Association and	NA

¹⁹ https://www.ilo.org/dyn/natlex/natlex4.detail?p_lang=en&p_isn=44862&p_country=MWI&p_classification=22

labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for		Protection of the Right to organize convention in 1999²⁰. The project participants will employ all workers in accordance with all applicable national laws. 2. Malawi ratified ILO N°98 Right to organise and collective bargaining convention in 1965. The project participants will not restrict any workers from establishing and joining labour organisations.	
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²⁰ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103101

payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and		3. Malawi ratified ILO N°29 Forced Labour Convention in 1999. All permanent workers will be provided with individual work agreements, including working hours, description of duties and tasks, remuneration, health insurance, termination of the contract, annual leave. 4. Malawi ratified ILO N°182 Worst Form of Child Labour Convention and ILO N°138 Minimum Age Convention in 1999. The project participants	
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emergency preparedness and response measures		do not engage in any form of child labour. The project participants will provide safe working environment, machinery and appropriate equipment. Training will be carried out for the production of stoves and health and safety measures. Accidents and incidents will be monitored and reported.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	No risks are foreseen in terms of negative consequences for local economy. The project will be financially sustainable through the sale of cookstoves and carbon finance. The project will also create jobs for local people.	NA
>>			
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project will not lead to any increase in greenhouse gas emissions. The project stoves will rather reduce	NA
>>			

		emissions due to the increased thermal efficiency compared to the baseline stoves.	
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project does not utilize any form of energy supply that is also being used by other users.	NA
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project does not use any water.	NA
>>			
Principle 8.2 Erosion and/or Water Body Instability			

a. Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?	No	The project includes installation of efficient cookstoves in households. No damage is foreseen for nature of soil or water bodies.	NA
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project includes installation of efficient cookstoves in households. No use of land or soil is applicable.	NA
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project includes installation of efficient cookstoves in households. No such risk is foreseen.	NA
>>			

Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	Not relevant	NA
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project includes installation of efficient cookstoves in households. Indoor air pollution in houses will be reduced compared to the baseline stoves.	NA
>>			
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	No	Not relevant	NA
>>			

Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	Not relevant	NA
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project aims to reduce firewood consumption by installing highly efficient cookstoves. This will reduce the harvest rate of forests.	NA
>>			
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	Not relevant	NA
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	Not relevant	NA
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	Implementation of the project will have a positive impact on forests; thus, to conservation of HCV ecosystems, critical habitats, landscapes, key biodiversity areas or other sites, by reducing the firewood harvesting.	Not relevant as impact is positive
>>			
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)?	No	Not relevant.	NA
b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?			
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	C-Quest Capital Stoves Asia Limited
Registration number with relevant authority	Brumby Centre, Lot 42, Jalan Muhibbah
Street/P.O. Box	
Building	Labuan F.T.
City	
State/Region	
Postcode	8700
Country	Malaysia
Telephone	+6 087 593828
E-mail	cqc-operations@cquestcapital.com
Website	https://cquestcapital.com
Contact person	Ken Newcombe
Title	Director
Salutation	Dr.
Last name	Newcombe
Middle name	
First name	Ken
Department	
Mobile	
Direct tel.	cqc-operations@cquestcapital.com
Personal e-mail	C-Quest Capital Stoves Asia Limited

APPENDIX 3- LUF ADDITIONAL INFORMATION

Risk of change to the Project Area during Project Certification Period:	N/A
Risk of change to the Project activities during Project Certification Period:	N/A
Land-use history and current status of Project Area:	N/A
Socio-Economic history:	N/A
Forest management applied (past and future)	N/A
Forest characteristics (including main tree species planted)	N/A
Main social impacts (risks and benefits)	N/A
Main environmental impacts (risks and benefits)	N/A
Financial structure	N/A
Infrastructure (roads/houses etc):	N/A
Water bodies:	N/A
Sites with special significance for indigenous people and local communities - resulting from the Stakeholder Consultation:	N/A
Where indigenous people and local communities are situated:	N/A
Where indigenous people and local communities have legal rights, customary rights or sites with special cultural, ecological, economic, religious or spiritual significance:	N/A

APPENDIX 4 - DESIGN CHANGES

A4.1. Details of proposed or actual design change

>>

N/A

A4.2. Describe the Impacts of Design Change on the following

a. Additionality

>>

N/A

b. Applicability of methodology and other methodological regulatory documents with which the project activity has been certified

>>

N/A

c. Compliance with the monitoring plan of the applied methodology

>>

N/A

d. Level of accuracy and completeness in the monitoring of the project activity compared with the requirements contained in the registered monitoring plan

>>

N/A

e. Scale of the project activity

>>

N/A

f. Stakeholder consultation

>>

N/A

g. Sustainable development criteria

>>

N/A

h. Safeguarding Assessment

>>

N/A

i. Compliance with applicable legislation

>>

N/A

Revision History

Version	Date	Remarks
2.1	14 April 2023	Integrated the design change memo as annex of the document.
2.0	4 May 2022	
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption